

Probability

Chapter 14 - Hinglish Notes, Formula, Examples + Solutions

Ye Chapter 14 (Probability) ka complete guide hai. Pehle red-page ke core basics (probability kya hai, experiment/outcome/event, favourable/total, deck of cards), fir chapter ka main content - formula, complement, coin/dice/cards ke problems. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Probability Kya Hai?

EXAM * Exam me CONSTANTLY aata hai

- Probability = kisi cheez ke hone ki sambhavna (chance).
- Hamesha 0 se 1 ke beech hoti hai.

2. Experiment, Outcome, Event

EXAM * Exam me CONSTANTLY aata hai

- Experiment = jo karte ho (coin uchhalna).
- Outcome = ek result (Head ya Tail).
- Event = outcomes ka group (jaise 'even number aaye').

3. Favourable Outcomes

EXAM * Exam me CONSTANTLY aata hai

- Favourable = jo hum chahte hain wale outcomes ki ginti.

4. Total Outcomes (Saare Possible)

EXAM * Exam me CONSTANTLY aata hai

- Coin: 2 (H, T). Dice: 6 (1-6).
- Deck of cards: 52.

5. Fraction

EXAM * Exam me CONSTANTLY aata hai

- Probability ek fraction hota hai: favourable / total.
- Jaise $\frac{3}{6} = \frac{1}{2}$.

6. Deck of Cards (52)

EXAM * Exam me CONSTANTLY aata hai

- 4 suits: Hearts, Diamonds (laal); Spades, Clubs (kaali).
- Har suit 13 cards. Face cards = J, Q, K (12 total).
- Aces = 4. Red = 26, Black = 26.

7. Sure vs Impossible

EXAM * Exam me CONSTANTLY aata hai

- Sure event -> probability 1.
- Impossible event -> probability 0.

PART B - Chapter 14 ka Main Content

8. Probability ka Formula

EXAM * Exam me CONSTANTLY aata hai

$$P(E) = (\text{Favourable outcomes}) / (\text{Total outcomes})$$

$$0 \leq P(E) \leq 1 \text{ (hamesha is range me).}$$

9. Complement of an Event

EXAM * Exam me CONSTANTLY aata hai

$$P(\text{not } E) = 1 - P(E)$$

$$P(E) + P(\text{not } E) = 1.$$

- Example: $P(\text{head}) = 1/2 \rightarrow P(\text{not head}) = 1 - 1/2 = 1/2$.

10. Coin Problems

EXAM * Exam me CONSTANTLY aata hai

- 1 coin: outcomes {H, T} -> total 2.
- 2 coins: {HH, HT, TH, TT} -> total 4.
- $P(\text{at least 1 head, 2 coins}) = 3/4$.

11. Dice Problems

EXAM * Exam me CONSTANTLY aata hai

- Total outcomes = 6 (1,2,3,4,5,6).
- Even {2,4,6} -> $P = 3/6 = 1/2$.
- Prime {2,3,5} -> $P = 3/6 = 1/2$.

12. Card Problems (52 cards)

EXAM * Exam me CONSTANTLY aata hai

- $P(\text{a King}) = 4/52 = 1/13$.
- $P(\text{red card}) = 26/52 = 1/2$.
- $P(\text{face card}) = 12/52 = 3/13$.

Quick Revision

$P(E) = \text{favourable} / \text{total}.$

$0 \leq P(E) \leq 1.$ Sure=1, Impossible=0.

$P(\text{not } E) = 1 - P(E).$

Coin=2, Dice=6, Cards=52 outcomes.

King=4/52=1/13, Red=1/2, Face=12/52=3/13.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1: Ek coin uchhala. P(head)?

- Step: Favourable (head) = 1, Total = 2.
- Step: $P = 1/2.$

Answer:1/2

Example 2: Ek dice fenka. P(even number)?

- Step: Even = {2,4,6} -> 3 favourable. Total = 6.
- Step: $P = 3/6 = 1/2.$

Answer:1/2

Example 3: Ek dice. P(number > 4)?

- Step: {5, 6} -> 2 favourable. Total = 6.
- Step: $P = 2/6 = 1/3.$

Answer:1/3

Example 4: 52 cards me se P(a King)?

- Step: Kings = 4. Total = 52.
- Step: $P = 4/52 = 1/13.$

Answer:1/13

Example 5: 52 cards. P(red card)?

- Step: Red = 26. Total = 52.
- Step: $P = 26/52 = 1/2.$

Answer:1/2

Example 6: P(head) = 1/2. P(not head)?

- Step: $P(\text{not } E) = 1 - P(E) = 1 - 1/2.$

Answer:1/2

Example 7: Dice. P(prime number)?

- Step: Primes = {2,3,5} -> 3. Total = 6.
- Step: $P = 3/6 = 1/2.$

Answer:1/2

Example 8: Bag me 3 red, 2 blue balls. P(red)?

- Step: Favourable = 3, Total = 3+2 = 5.
- Step: $P = 3/5.$

Answer:3/5

Example 9: 52 cards. P(face card)?

- Step: Face cards = J,Q,K x 4 suits = 12. Total = 52.
- Step: $P = 12/52 = 3/13.$

Answer:3/13

Example 10: 2 coins. P(at least one head)?

- Step: Outcomes: HH, HT, TH, TT (total 4).
- Step: At least 1 head = HH, HT, TH = 3.
- Step: $P = 3/4$.

Answer:3/4

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1.** Probability ki value kis range me hoti hai?
- Q2.** Probability ka formula kya hai?
- Q3.** Sure event ki probability kitni hoti hai?
- Q4.** Impossible event ki probability kitni hoti hai?
- Q5.** $P(\text{not } E)$ ka formula kya hai?
- Q6.** Ek coin me total outcomes kitne hote hain?
- Q7.** Ek dice me total outcomes kitne hote hain?
- Q8.** Deck me total cards kitne hote hain?
- Q9.** Ek coin. $P(\text{tail})$?
- Q10.** Ek dice. $P(\text{odd number})$?
- Q11.** Ek dice. $P(\text{number } 3)$?
- Q12.** 52 cards. $P(\text{a Queen})$?
- Q13.** 52 cards. $P(\text{black card})$?
- Q14.** 52 cards. $P(\text{an Ace})$?
- Q15.** Bag me 4 red, 6 green. $P(\text{green})$?
- Q16.** Dice. $P(\text{number} \leq 2)$?
- Q17.** $P(E)=2/5$ ho to $P(\text{not } E)$?
- Q18.** 52 cards me face cards kitne hote hain?
- Q19.** 2 coins. Total outcomes kitne?
- Q20.** Ek dice. $P(\text{multiple of } 3)$?

Step-by-Step Solutions

Q1. Probability range?

- Step: 0 se 1.

Answer: $0 \leq P \leq 1$

Q2. Probability formula?

- Step: Standard.

Answer:favourable / total

Q3. Sure event probability?

- Step: Pakka hoga.

Answer:1

Q4. Impossible event probability?

- Step: Kabhi nahi hoga.

Answer:0

Q5. $P(\text{not } E)$ formula?

- Step: Complement.

Answer: $1 - P(E)$

Q6. Coin total outcomes?

- Step: H, T.

Answer:2

Q7. Dice total outcomes?

- Step: 1 se 6.

Answer:6

Q8. Deck total cards?

- Step: Standard deck.

Answer:52

Q9. Coin $P(\text{tail})$?

- Step: 1 favourable / 2.

Answer: $1/2$

Q10. Dice $P(\text{odd})$?

- Step: $\{1, 3, 5\} = 3/6$.

Answer: $1/2$

Q11. Dice $P(3)$?

- Step: 1 favourable / 6.

Answer: $1/6$

Q12. 52 cards $P(\text{Queen})$?

- Step: $4/52$.

Answer: $1/13$

Q13. 52 cards $P(\text{black})$?

- Step: $26/52$.

Answer: $1/2$

Q14. 52 cards $P(\text{Ace})$?

- Step: $4/52$.

Answer: $1/13$

Q15. 4 red, 6 green $P(\text{green})$?

- Step: $6/(4+6) = 6/10$.

Answer: $3/5$

Q16. Dice $P(\leq 2)$?

- Step: $\{1, 2\} = 2/6$.

Answer: $1/3$

Q17. $P(E) = 2/5 \rightarrow P(\text{not } E)$?

- Step: $1 - 2/5$.

Answer: $3/5$

Q18. Face cards count?

- Step: J,Q,K x 4.

Answer:12

Q19. 2 coins total outcomes?

- Step: HH,HT,TH,TT.

Answer:4

Q20. Dice P(multiple of 3)?

- Step: {3,6} = 2/6.

Answer:1/3